

Exercise Solutions

1. User Upcoming Events

```
SELECT u.full_name,e.title,e.start_date FROM Users u JOIN Registrations r  
ON u.user_id=r.user_id JOIN Events e ON e.event_id=r.event_id WHERE  
e.status='upcoming' AND u.city=e.city ORDER BY e.start_date;
```

2. Top Rated Events

```
SELECT e.event_id,e.title,AVG(f.rating) avg_rating FROM Events e JOIN  
Feedback f ON e.event_id=f.event_id GROUP BY e.event_id,e.title HAVING  
COUNT(*)>=10 ORDER BY avg_rating DESC;
```

3. Inactive Users

```
SELECT * FROM Users u WHERE NOT EXISTS (SELECT 1 FROM Registrations r  
WHERE r.user_id=u.user_id AND r.registration_date>=CURDATE()-  
INTERVAL 90 DAY);
```

4. Peak Session Hours

```
SELECT event_id,COUNT(*) total_sessions FROM Sessions WHERE  
TIME(start_time) BETWEEN '10:00:00' AND '12:00:00' GROUP BY event_id;
```

5. Most Active Cities

```
SELECT u.city,COUNT(DISTINCT r.user_id) registrations FROM Users u JOIN  
Registrations r ON u.user_id=r.user_id GROUP BY u.city ORDER BY  
registrations DESC LIMIT 5;
```

6. Event Resource Summary

```
SELECT e.title,COUNT(*) resources FROM Events e LEFT JOIN Resources r  
ON e.event_id=r.event_id GROUP BY e.event_id,e.title;
```

7. Low Feedback Alerts

```
SELECT u.full_name,f.comments,e.title FROM Feedback f JOIN Users u ON  
u.user_id=f.user_id JOIN Events e ON e.event_id=f.event_id WHERE  
f.rating<3;
```

8. Sessions per Upcoming Event

```
SELECT e.title,COUNT(s.session_id) session_count FROM Events e LEFT JOIN  
Sessions s ON e.event_id=s.event_id WHERE e.status='upcoming' GROUP  
BY e.event_id,e.title;
```

9. Organizer Event Summary

```
SELECT organizer_id,status,COUNT(*) total_events FROM Events GROUP BY  
organizer_id,status;
```

10. Feedback Gap

```
SELECT e.title FROM Events e JOIN Registrations r ON e.event_id=r.event_id  
LEFT JOIN Feedback f ON e.event_id=f.event_id GROUP BY  
e.event_id,e.title HAVING COUNT(f.feedback_id)=0;
```

11. Daily New User Count

```
SELECT registration_date,COUNT(*) users FROM Users GROUP BY  
registration_date;
```

12. Event with Maximum Sessions

```
SELECT event_id,COUNT(*) cnt FROM Sessions GROUP BY event_id HAVING  
cnt=(SELECT MAX(c) FROM (SELECT COUNT(*) c FROM Sessions GROUP BY  
event_id)x);
```

13. Average Rating per City

```
SELECT e.city,AVG(f.rating) avg_rating FROM Events e JOIN Feedback f ON  
e.event_id=f.event_id GROUP BY e.city;
```

14. Most Registered Events

```
SELECT e.title,COUNT(*) registrations FROM Events e JOIN Registrations r
ON e.event_id=r.event_id GROUP BY e.event_id,e.title ORDER BY
registrations DESC LIMIT 3;
```

15. Event Session Time Conflict

```
SELECT s1.event_id,s1.session_id,s2.session_id FROM Sessions s1 JOIN
Sessions s2 ON s1.event_id=s2.event_id AND s1.session_id<s2.session_id
AND s1.start_time<s2.end_time AND s2.start_time<s1.end_time;
```

16. Unregistered Active Users

```
SELECT * FROM Users u WHERE u.registration_date>=CURDATE()-INTERVAL
30 DAY AND NOT EXISTS(SELECT 1 FROM Registrations r WHERE
r.user_id=u.user_id);
```

17. Multi-Session Speakers

```
SELECT speaker_name,COUNT(*) sessions FROM Sessions GROUP BY
speaker_name HAVING COUNT(*)>1;
```

18. Resource Availability Check

```
SELECT e.title FROM Events e LEFT JOIN Resources r ON
e.event_id=r.event_id WHERE r.resource_id IS NULL;
```

19. Completed Events with Feedback Summary

```
SELECT e.title,COUNT(DISTINCT r.registration_id) registrations,AVG(f.rating)
avg_rating FROM Events e LEFT JOIN Registrations r ON
e.event_id=r.event_id LEFT JOIN Feedback f ON e.event_id=f.event_id
WHERE e.status='completed' GROUP BY e.event_id,e.title;
```

20. User Engagement Index

```
SELECT u.full_name,COUNT(DISTINCT r.event_id)
attended,COUNT(DISTINCT f.feedback_id) feedbacks FROM Users u LEFT
```

```
JOIN Registrations r ON u.user_id=r.user_id LEFT JOIN Feedback f ON  
u.user_id=f.user_id GROUP BY u.user_id,u.full_name;
```

21. Top Feedback Providers

```
SELECT u.full_name,COUNT(*) feedbacks FROM Users u JOIN Feedback f  
ON u.user_id=f.user_id GROUP BY u.user_id,u.full_name ORDER BY  
feedbacks DESC LIMIT 5;
```

22. Duplicate Registrations Check

```
SELECT user_id,event_id,COUNT(*) cnt FROM Registrations GROUP BY  
user_id,event_id HAVING COUNT(*)>1;
```

23. Registration Trends

```
SELECT YEAR(registration_date) yr,MONTH(registration_date)  
mn,COUNT(*) cnt FROM Registrations GROUP BY  
YEAR(registration_date),MONTH(registration_date) ORDER BY yr,mn;
```

24. Average Session Duration per Event

```
SELECT event_id,AVG(TIMESTAMPDIFF(MINUTE,start_time,end_time))  
avg_minutes FROM Sessions GROUP BY event_id;
```

25. Events Without Sessions

```
SELECT e.title FROM Events e LEFT JOIN Sessions s ON  
e.event_id=s.event_id WHERE s.session_id IS NULL;
```